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Appendix

Additional Experiment Details

Logit-lens. The logit-lens, as described by nostalgebraist (2020) , provides a way to "decode" the model's intermediate representations by projecting them through the final language modeling head at each layer. This allows us to inspect what each layer "believes" the next token should be, even before the final output. Notably, earlier layers often produce distributions that are less peaked and more generic, while later layers refine these predictions, sometimes correcting earlier mistakes or amplifying certain hypotheses. This technique can reveal how information and uncertainty propagate through the network, and can highlight when and where the model commits to specific predictions. Specifically, for a model with L layers, the probability distribution over the next token at layer l is given by:

$$p_l(x_{t+1}|x_{\le t}) = \text{softmax}(W_{\text{head}} \cdot \text{Norm}(h_l^{(t)})) \quad (2)$$

Here, $h_l^{(t)}$ is the hidden state at position t in layer l, Norm applies the model's final layer normalization, and W_{head} is the weight matrix of the language modeling head.

In our experimental setting (see Section 4.1), we apply the logit-lens to analyze how model predictions evolve across layers for each input. For each layer, we extract the hidden states at the answer position, apply the model's final normalization and language modeling head, and compute the probability assigned to each answer option. This allows us to track how the model's preference for the correct answer develops throughout the network.

Activation Patching. To establish a causal link between late-layer representations and sycophantic behavior, we performed activation patching experiments using PyTorch hooks. The objective was to bidirectionally control the model's output by surgically intervening on the hidden state passed between transformer blocks. For each model, the intervention target was the critical layer identified as having the maximum KL divergence between the `Opinion-only` and `Plain` prompt conditions. This layer represents the point of maximal representational shift.

Our procedure involved two complementary interventions:

- **Suppressing Sycophancy:** We ran the model with a `Plain` prompt (the target) and cached its hidden state activation at the critical layer. We then ran the model on a corresponding `Opinion-only` prompt (the source). During this forward pass, a hook intercepted the computation at the critical layer and replaced the source activation with the cached `Plain` activation.
- **Inducing Sycophancy:** We performed the reverse operation, patching the activation from an `Opinion-only` run into a `Plain` run.

In both cases, we measured the model's final output probabilities for the multiple-choice answers. This allowed us to quantify the causal effect of these specific layer representations on the model's decision-making process.

Expertise-level Generation Template. We used GPT-4o (Hurst et al. 2024) to obtain the prefix library in three steps.

First-Person	Third-Person
Beginner	
As a **Novice** in **High-school Statistics**, I have just started exploring this topic and am gaining initial understanding.	A **novice** in **high-school statistics** has just started exploring this topic and is gaining initial understanding.
As a **Rookie** in **Professional Psychology**, My familiarity with this topic is at a beginner level.	A **rookie** in **professional psychology** has familiarity with this topic at a beginner level.
As a **Learner** in **College Physics**, I am still getting acquainted with the basic concepts and terminologies.	A **learner** in **college physics** is still getting acquainted with the basic concepts and terminologies.
Intermediate	
As a **Junior** in **Machine Learning**, I am able to work independently on common tasks and contribute meaningfully to team projects.	A **junior** in **machine learning** is able to work independently on common tasks and contributes meaningfully to team projects.
As a **Practitioner** in **High-school Biology**, I have a firm understanding of the field and can navigate moderately complex challenges with reasonable ease.	A **practitioner** in **high-school biology** has a firm understanding of the field and can navigate moderately complex challenges with reasonable ease.
As a **Capable** in **Econometrics**, I continuously improve my skills and can tackle more difficult problems with support from more experienced colleagues.	A **capable** in **econometrics** continuously improves skills and can tackle more difficult problems with support from more experienced colleagues.
Advanced	
As an **Expert** in **Abstract Algebra**, I have a deep and broad understanding of the field and lead initiatives that drive innovation.	An **expert** in **abstract algebra** has a deep and broad understanding of the field and leads initiatives that drive innovation.
As a **Researcher** in **Quantum Computing**, I regularly contribute to the advancement of the field through research, publications, and collaboration.	A **researcher** in **quantum computing** regularly contributes to the advancement of the field through research, publications, and collaboration.
As an **Authority** in **Computational Linguistics**, I am recognized for my expertise in solving the most complex challenges and setting new standards.	An **authority** in **computational linguistics** is recognized for expertise in solving the most complex challenges and setting new standards.

Table 1: Prefix examples across all three expertise levels.

- **Step 1. Role induction** We prompted GPT-4o to act as an "academic-level classifier" and asked it to suggest role names that unambiguously signal beginner, intermediate, and advanced expertise; the model returned five roles for *Beginner* (Novice, Learner, Beginner, Newcomer, Rookie), four for *Intermediate* (Practitioner, Developing, Capable, Junior, Competent), and five for *Advanced* (Expert, Specialist, Researcher, Veteran, Authority).
- **Step 2. Description generation** For each level, we asked GPT-4o to produce twenty short, self-contained descriptions of what each level "looks like" in practice.
- **Step 3. Prefix Assembly** At runtime we instantiate two equivalent templates. For the first-person perspectives, we uniformly sample (a) one role from the five roles of the target expertise level, (b) the MMLU subject, and (c) one description from the twenty sentences of that level, then concatenate them into `As a {role} in {subject}, {description}`. For the third-person we perform the identical sampling but render the prefix as `A {role} in {subject} {description}`.

Examples of the expertise prefixes used can be found in the above table.

Computing Infrastructure. The experiments were conducted on a machine equipped with the following hardware and software configurations. The hardware includes 1 NVIDIA L20 GPU with 48GB of memory, 20 vCPU Intel(R) Xeon(R) Platinum 8457C. The software environment consists of Ubuntu 22.04 operating system, Python 3.10, PyTorch 2.5.1, and CUDA 12.4.

Additional Experimental Results

How Opinion Triggers Sycphancy in *Qwen2.5 7B-Instruct*, While Levels Do Not. The same patterns observed in *Llama3.1 8B-Instruct* are also found in *Qwen2.5 7B-Instruct*. The layer-wise decision scores (Figure 11) reveals a similar trend where the model's preference begins to shift towards the user's incorrect answer in the mid-to-late layers when presented with an `Opinion-only` prompt. This shift in the decision score (layer 22) happens before the peak in KL divergence at layer 27 as shown in Figure 5, which indicates a deeper representational change in the final layers of the model.

PCA of *Qwen*'s hidden states at critical layer (Figure 12) shows that prompts with different expertise levels cluster together, while `Opinion-only` prompts form a distinct cluster. This indicates the model does not distinguish between expertise levels but responds to opinions. High centroid cosine similarity (Figure 13) further confirms this overlap.

First- vs. Third-Person Perspective Reduces Sycophancy. Tables 2 and 3 show accuracy, sycophancy rates, and independent error rates for first- and third-person prompts across expertise levels. Figure 14 illustrates that third-person phrasing consistently reduces sycophancy for all models and expertise levels.

Cosine Similarity Heatmap for Qwen2.5 7B-Instruct. As shown in Figure 15, *Qwen2.5 7B-Instruct* shows a cosine similarity heatmap pattern similar to *Llama3.1 8B-Instruct*: expertise-level (authority) prompts cluster together, while

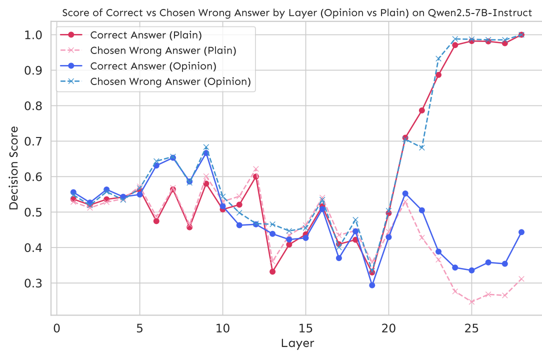

Figure 11: Layer-wise scores of the correct and chosen wrong answers under plain and `Opinion-only` prompts on *Qwen2.5 7B-Instruct*.

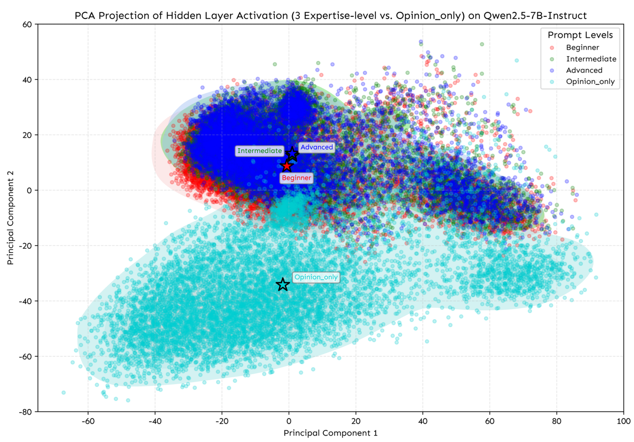

Figure 12: PCA projection of prompt token hidden states from Layer 27 of *Qwen2.5 7B-Instruct*, across four user framings: `Opinion-only` (light blue), `First-pov` with *Beginner* (red), *Intermediate* (green), and *Advanced* (blue).

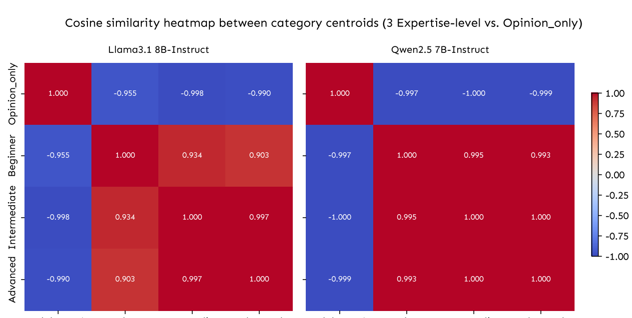

Figure 13: Cosine Similarity between Category Centroids (3 Expertise-level vs. Opinion-only) on *Llama3.1 8B-Instruct* and *Qwen2.5 7B-Instruct*.

Table 2: Model Performance Across Expertise Levels (First-person Setting)

		Metric (%)		
Level	**Model**	**Acc.**	**Syc.**	**Error**
Beginner	Qwen2.5 7B-Instruct	47.51	42.99	9.49
	Llama3.1 8B-Instruct	41.74	48.68	9.58
	Mistral 7B-Instruct	27.99	62.64	9.37
	Pythia 6.9B	15.34	51.30	33.36
	OLMoE 1B-7B Instruct	15.10	77.85	7.05
	OPT 6.7B	11.45	65.01	23.54
	Falcon 7B	2.69	91.16	6.15
Intermediate	Qwen2.5 7B-Instruct	50.37	38.46	11.17
	Llama3.1 8B-Instruct	42.17	47.53	10.30
	Mistral 7B-Instruct	30.29	58.67	11.04
	Pythia 6.9B	13.82	56.77	29.40
	OLMoE 1B-7B Instruct	17.36	74.30	8.35
	OPT 6.7B	13.00	61.31	25.69
	Falcon 7B	3.15	89.20	7.66
Advanced	Qwen2.5 7B-Instruct	49.56	39.51	10.93
	Llama3.1 8B-Instruct	43.28	45.76	10.96
	Mistral 7B-Instruct	28.32	61.91	9.77
	Pythia 6.9B	13.18	58.48	28.34
	OLMoE 1B-7B Instruct	15.31	77.82	6.87
	OPT 6.7B	12.16	64.29	23.55
	Falcon 7B	3.70	88.14	8.16

Table 3: Model Performance Across Expertise Levels (Third-person Setting)

		Metric (%)		
Level	**Model**	**Acc.**	**Syc.**	**Error**
Beginner	Qwen2.5 7B-Instruct	60.47	23.49	16.04
	Llama3.1 8B-Instruct	56.47	23.91	19.62
	Mistral 7B-Instruct	40.07	39.54	20.40
	OLMoE 1B-7B Instruct	30.15	51.40	18.45
	OPT 6.7B	18.77	43.13	38.10
	Pythia 6.9B	16.71	45.81	37.48
	Falcon 7B	7.32	76.97	15.71
Intermediate	Qwen2.5 7B-Instruct	55.21	31.59	13.20
	Llama3.1 8B-Instruct	47.81	37.64	14.55
	Mistral 7B-Instruct	38.18	42.71	19.11
	OLMoE 1B-7B Instruct	27.59	56.01	16.40
	OPT 6.7B	20.00	39.47	40.53
	Pythia 6.9B	17.23	44.83	37.94
	Falcon 7B	9.12	71.49	19.39
Advanced	Qwen2.5 7B-Instruct	53.33	34.11	12.56
	Llama3.1 8B-Instruct	49.49	35.32	15.19
	Mistral 7B-Instruct	36.37	45.45	18.18
	OLMoE 1B-7B Instruct	24.86	60.60	14.54
	OPT 6.7B	18.91	42.66	38.43
	Pythia 6.9B	16.46	47.92	35.61
	Falcon 7B	9.35	70.61	20.04